

Inline and Display Mathematics

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\frac{1}{4!} \int_{\Lambda} \frac{d^4 \vec{p}_1}{(2\pi)^4} \frac{d^4 \vec{p}_2}{(2\pi)^4} \frac{d^4 \vec{p}_3}{(2\pi)^4} \frac{d^4 \vec{p}_5}{(2\pi)^4} \delta \lambda_4(1235) \delta^4(\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \vec{p}_5) \hat{\phi}_2(\vec{p}_1) \hat{\phi}_2(\vec{p}_2) \hat{\phi}_2(\vec{p}_3) \hat{\phi}_2(\vec{p}_5).$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\frac{\partial T^{ab}}{\partial w_{ij}} = \frac{\partial T^{ab}}{\partial w_{ji}}, \frac{\partial T^{ab}}{\partial w_{ijk}} = \frac{\partial T^{ab}}{\partial w_{jik}}, \frac{\partial T^{ab}}{\partial w_{ijkl}} = \frac{\partial T^{ab}}{\partial w_{jikl}} = \frac{\partial T^{ab}}{\partial w_{ijlk}} = \frac{\partial T^{ab}}{\partial w_{jilk}}.$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\mathcal{W}^{\text{U}(1)} = \partial_{\mu} \frac{\delta}{\delta A_{\mu}} + \varepsilon^{ab} \left(A_{\mu}^a \frac{\delta}{\delta A_{\mu}^b} + c^a \frac{\delta}{\delta c^b} + b^a \frac{\delta}{\delta b^b} + \bar{c}^a \frac{\delta}{\delta \bar{c}^b} + A_{\mu}^{a*} \frac{\delta}{\delta A_{\mu}^{b*}} + c^{a*} \frac{\delta}{\delta c^{b*}} \right).$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$q^{\frac{1}{2}\mathcal{E}(d)} \sum_{(d^{\bullet}; d^{\infty})} (-1)^{l(d^{\bullet})} q^{-\chi(d^{\bullet}) - \chi(d^{\infty}, d^{\bullet}) - \mathcal{E}(d^{\bullet})} \left(\prod_{i=1}^{l(d^{\bullet})} \frac{q^{-\chi(d^i, d^i)}}{(q^{-1})_{d^i}} \right) \frac{q^{-\mathcal{E}(d^{\infty})}}{(q^{-1})_{d^{\infty}}^{\sigma}},$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$\phi_R \rightarrow B' e^{-\gamma ax} e^{\gamma aL} \frac{\Gamma(2\gamma)\Gamma(1+2\sigma)}{\Gamma(1+\gamma+\sigma+\lambda)\Gamma(\gamma+\sigma-\lambda)} + B' e^{\gamma ax} e^{-\gamma aL} \frac{\Gamma(-2\gamma)\Gamma(1+2\sigma)}{\Gamma(1-\gamma+\sigma-\lambda)\Gamma(-\gamma+\sigma-\lambda)}$$